

Bharath Chirra

Data Analyst

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SUMMARY

- Data Analyst with **4+ years of experience** across **healthcare, finance, and consulting domains**, skilled in building data infrastructure, ensuring **HIPAA** compliance, and delivering insights that enhanced process efficiency and compliance tracking.
- Engineered **30+ ETL workflows** using **Apache Airflow, AWS Glue, and Informatica**, consolidating data from multiple sources and reducing manual processing by **10+ hours weekly**.
- Created executive-level dashboards in **Power BI, Tableau, and Excel**, enabling real-time monitoring of KPIs, claim trends, and compliance metrics.
- Applied **machine learning techniques, A/B testing, and forecast modeling** using **Python, R, and TensorFlow**, driving audit automation and improving resource planning.
- Experienced in designing anomaly detection pipelines and integrating **Kafka streams**, enhancing trade operations and regulatory reporting in financial environments.
- Optimized QA cycles using **PyTest, Postman, and Selenium**, reducing defect leakage and improving test traceability across reporting platforms.
- Delivered cloud-native analytics solutions with **AWS, Docker, and CI/CD pipelines**, ensuring scalable deployment across development, test, and production.

PROFESSIONAL EXPERIENCE

Data Analyst | Blue Cross Blue Shield | INDIANA, USA

Apr 2025 – Current

- Constructed **claims ingestion framework** using **Apache Airflow** and **AWS Glue**, automating data intake from 5 internal systems and saving **10+ manual hours weekly**.
- Delivered **Power BI dashboards** visualizing **KPIs** in **enrollment, denials, and provider usage**, supporting **operational reviews** across 3 departments.
- Executed **anomaly detection** on **1.2M+ transactions** using **SQL** and **Pandas**, uncovering **28 compliance violations** flagged by audit.
- Designed **A/B testing** experiments in **Python** and **R** to measure **policy engagement**, comparing 3 intervention strategies.
- Defined and standardized **15+ metrics** in **Snowflake** for **actuarial benchmarking**, enabling **cost-efficiency comparisons**.
- Created **test automation** using **PyTest** and **Postman** for **claims APIs**, improving **test traceability** and cutting **QA rework**.
- Deployed **analytics scripts** via **Docker** and **GitHub Actions**, enabling report execution across **development, test, and production**.
- Partnered with cross-functional teams (operations, clinical, compliance) to align dashboards with **HIPAA** privacy standards.
- Produced **time series forecasts** using **scikit-learn** and **TensorFlow** to predict **volume shifts** and improve **resource alignment**.

Data Analyst | BNY Mellon | INDIANA, USA

Aug 2024 – Apr 2025

- Supported **ETL architecture** using **Informatica** and **SQL**, transforming multi-source data into **centralized data warehouses**.
- Constructed **7 interactive Tableau dashboards** for tracking **portfolio exposure, risk posture, and regulatory indicators**.
- Enhanced **workflow efficiency** in **AWS Glue** and **Apache Airflow**, cutting pipeline runtime by **4 hours per day**.
- Performed **statistical modeling** and **EDA** on **500K+ transactions**, uncovering **14 hidden trends** linked to payment delays.
- Streamlined **real-time data integration** using **Kafka** and custom API connectors, triggering **5 proactive alert systems**.
- Authored **20+ advanced SQL procedures** across **PostgreSQL** and **Snowflake** for financial oversight.
- Automated **quality checks** with **PyTest** and **Selenium**, improving test coverage and minimizing manual verification.
- Scheduled production-ready tasks using **Docker** and **Jenkins**, achieving a **6-hour weekly savings** in execution time.

Data Analyst | KPMG | India

Mar 2020 – Aug 2022

- Managed **ETL workloads** via **Informatica** and **AWS Glue**, harmonizing data from 6 operational departments for audit reports.
- Delivered SQL-based **compliance reports** using **MySQL** and **Snowflake**, detecting discrepancies in transactional data.
- Built **5 Tableau dashboards** aligning with internal risk control frameworks across key business units.
- Cleaned and transformed **800K+ raw records** using **Python**, correcting schema mismatches and null-value inconsistencies.
- Defined **12 organizational KPIs** with business stakeholders to measure service performance under regulatory guidelines.
- Automated weekly reporting through **Power BI** and **Excel macros**, eliminating **7+ hours** of manual data handling.
- Traced **20+ user onboarding inconsistencies** through **log reviews** and **data lineage mapping**.
- Participated in **Agile sprints** to verify analytics deliverables met audit, QA, and compliance requirements.

Data Analyst Intern | Adani Group | India

Sep 2019 – Feb 2020

- Contributed to ETL prep tasks using **SQL queries** and **Excel macros** from 3 internal platforms for daily analytics operations.
- Evaluated **250K+ sensor logs** with **Python (Pandas, NumPy)** to highlight **18 recurring electrical load anomalies**.
- Created real-time **Excel dashboards** to monitor distribution trends and power unit performance metrics.
- Designed SQL scripts to extract and merge **SCADA system data** with billing records for consolidated reporting.
- Generated **4 interactive Power BI dashboards** visualizing utility patterns, aiding internal audit workflows.
- Modeled **energy KPIs** by correlating output with usage costs across 5 plant locations.
- Conducted **time-series EDA** to identify misalignments in consumption forecasts and suggest operational corrections.
- Documented code scripts, data models, and presentation slides for final internship evaluation and knowledge sharing.

PROJECTS

Interactive Campus Map

- Developed a full-stack **campus navigation web app** using **React** for the frontend and **Node.js** for the backend.
- Integrated **Google Maps API**, **React Auto-complete**, and **interactive map pins** for accurate indoor navigation across 75+ campus locations.
- Built a **React Native geolocation microservice** to support real-time indoor positioning within campus buildings.

Real-Time Intrusion Detection System

- Created a real-time **intrusion detection system** using **Django** (backend) and **React** (frontend), simulating IoT-based attack scenarios.
- Integrated **Arduino UNO**, **NodeMCU**, and **Ultrasonic sensors** with an **Apache server** to capture physical access attempts.
- Applied **machine learning models** with **Kafka** and **WebSockets** to classify and display 5+ simulated **cyber-attacks** in real time.

TECHNICAL SKILLS

- **Data Visualization & Reporting:** Power BI, Tableau, Matplotlib, Seaborn, Plotly, Excel (Power Pivot/Advanced Charts)
- **Programming Languages:** Python (Pandas, NumPy, Flask), Java (Spring Boot), SQL (Advanced Queries), R, JavaScript/TypeScript
- **Cloud & DevOps:** AWS(Redshift, Glue, Lambda), Terraform, Docker, CI/CD (Jenkins, GitHub Actions), Databricks
- **Database & Big Data:** SQL (MySQL/PostgreSQL), NoSQL (MongoDB/Cassandra), Spark, Elasticsearch, Snowflake
- **Machine Learning & AI:** Supervised Learning, LLMs (Hugging Face), TensorFlow, PyTorch, Scikit-learn, Time Series Forecasting
- **ETL & Data Engineering:** Apache Airflow, Informatica, AWS Glue, Data Warehousing (Star, Snowflake Schema)
- **Testing & QA:** Unit Testing (PyTest, JUnit), Selenium, Postman, Test Automation
- **Tools & Platforms:** Git, JIRA, Kafka, Linux/Unix, CRM (Salesforce), Google Analytics, HIPAA Compliance

EDUCATION

Master of Science in Computer Science Purdue University at Hammond, IN	<i>Aug 2022 – May 2024</i>
Bachelor of Science in Information Technology Vidya Jyothi Institute of Technology at Hyderabad, India	<i>Aug 2016 – Aug 2020</i>

CERTIFICATIONS

- **Oracle Certified Associate, Java SE 8 Programmer**
- **ISTQB Certified Agile Tester (CTFL-AT)**
- **ISTQB Certified Tester – Foundation Level (CTFL)**